An AI-Powered Crop Disease Detection and Agrochemical Recommendation Framework

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***Abstract*—Crop diseases are a major challenge in agriculture, as they can reduce crop productivity and affect the quality of agricultural output. Early and accurate disease identification is essential for supporting timely crop management decisions. Traditional disease diagnosis often relies on manual inspection, which can be time-consuming and may require specialized agricultural knowledge. This paper presents an AI-powered framework for crop disease detection and agrochemical recommendation using deep learning. The framework employs the EfficientNetB0 convolutional neural network, initialized with ImageNet-pretrained weights, to classify plant leaf images into disease categories. The model is trained and validated using the PlantVillage dataset, which contains labeled images of healthy and diseased plant leaves. Transfer learning is applied to improve feature extraction and classification performance. Following disease identification, the framework is designed to provide disease-specific agrochemical guidance through a recommendation module. The proposed approach aims to connect automated disease diagnosis with actionable agricultural information through an integrated system. The framework can support early disease identification and assist users in making informed crop-management decisions. Its effectiveness is evaluated using classification performance metrics, including accuracy, precision, recall, and F1-score. The work highlights the potential of deep learning-based crop disease detection integrated with recommendation support for precision agriculture.**

***Index Terms*—Crop disease detection, EfficientNetB0, deep learning, transfer learning, PlantVillage dataset, agrochemical recommendation, precision agriculture.**

I. Introduction

Agriculture plays a vital role in food security and economic development. However, crop diseases can significantly reduce agricultural productivity, affect crop quality, and increase production costs. Early disease detection is therefore essential for minimizing crop losses. Conventional diagnosis mainly relies on visual inspection, which can be time-consuming and requires domain expertise.

Recent advancements in deep learning have enabled automated plant disease detection using leaf images. Convolutional Neural Networks (CNNs) and architectures such as DenseNet, EfficientNet, and Vision Transformers have shown promising results in disease classification. The PlantVillage dataset is widely used for evaluating these methods. However, models trained on controlled images may face challenges in real-world conditions due to variations in lighting, backgrounds, and image quality.

Despite the progress in disease classification, identifying a disease alone may not provide sufficient guidance for effective crop management. Farmers also require appropriate treatment and prevention measures. Existing studies have explored pesticide and fertilizer recommendation systems, but integrating disease detection with reliable, disease-specific agrochemical guidance remains a challenge.

To address this gap, this paper proposes an AI-powered crop disease detection and agrochemical recommendation framework. The disease detection component uses EfficientNetB0 with transfer learning and is trained and validated using the PlantVillage dataset. The predicted disease category is then connected to an agrochemical recommendation module to support informed crop-management decisions.

The main contributions of this work are:

- Development of an EfficientNetB0-based crop disease classification component using transfer learning.
- Integration of disease classification with disease-specific agrochemical recommendation support.
- Evaluation of the disease detection model using accuracy, precision, recall, and F1-score.

The paper is organized as follows: Section II reviews related work, Sections III and IV describe the proposed framework and methodology, Section V presents the results, Section VI discusses agrochemical recommendations, and Section VII concludes the paper.

II. Related Work

Recent studies have explored deep learning techniques for automated plant disease detection. CNN-based architectures, transfer learning, and advanced models such as DenseNet and Vision Transformers have demonstrated promising results in classifying plant diseases using leaf images. The PlantVillage dataset is commonly used for evaluating these approaches. Researchers have also investigated explainable AI, disease severity estimation, and agricultural recommendation systems to improve decision-making. However, challenges such as variations in real-world conditions, computational complexity, and the integration of disease detection with reliable agrochemical recommendations remain. Table I presents a comparative analysis of selected studies, highlighting their methodologies, limitations, and relevance to the proposed framework.

TABLE I
LITERATURE REVIEW OF SELECTED EXISTING STUDIES

S.No.	Author, Conference, Year	Algorithm Used	Key Methodology	Limitation	Research Gap	How It Helps Our Project
1	L. Li et al., IEEE Access, 2021	Deep Learning, CNNs	Reviews deep learning approaches for plant disease detection and classification using image-based methods.	Requires large and diverse datasets; performance may decrease under real-field conditions.	Improved robustness and generalization to field images are needed.	Provides the foundation for using deep learning for automated crop disease classification.
2	A. Dolatabadian et al., Plant Pathology, 2025	ML, DL, CNNs	Reviews image-based disease detection using ML/DL, imaging sensors, and contextual data.	Lighting, background, and environmental variations affect performance.	More reliable detection under real agricultural conditions is needed.	Supports the need for robust image-based disease detection.
3	M. Majdalawieh et al., Smart Agricultural Technology, 2025	CNNs, Vision Transformers	Systematically reviews AI methods, datasets, and evaluation techniques for crop disease detection.	ViTs require high computational resources; dataset bias affects deployment.	Efficient and generalizable disease detection models are needed.	Supports the selection and evaluation of deep learning architectures.
4	A. Oad et al., IEEE Access, 2024	Ensemble Learning, XAI	Combines ensemble learning and explainable AI for plant leaf disease classification.	Higher computational complexity and dependence on quality training data.	Lightweight and interpretable models remain important.	Motivates future explainability and robust prediction extensions.
5	A. Bhargava et al., IEEE Access, 2024	Computer Vision, AI	Reviews computer vision and AI techniques for plant leaf disease detection, classification, and diagnosis.	Performance depends on dataset quality and real-field image variations.	Models need improved adaptability to field conditions.	Supports the use of image-based AI for crop disease diagnosis.
6	A. Al Mamun et al., IEEE Access, 2024	Self-Supervised Learning	Reviews self-supervised learning techniques for plant disease detection with limited labeled data.	Training complexity and real-field variations remain challenges.	Effective learning with limited labeled agricultural images is needed.	Highlights dataset limitations and possible future use of self-supervised learning.
7	S. Banothu et al., Cogent Engineering, 2024	DenseNet, Pesticide Recommendation	Uses DenseNet for plant disease identification and recommends pesticides based on the detected disease.	Performance may be affected by image quality and disease symptom variation.	Disease-specific recommendations require reliable classification.	Provides a direct reference for connecting disease detection with pesticide guidance.
8	N. Nigar et al., IEEE Access, 2024	Deep Learning, XAI	Combines deep learning-based plant disease classification with explainable AI.	Requires sufficient training data and computational resources.	Improved interpretability and field generalization are needed.	Supports deep learning-based classification and possible explainability extensions.
9	A. Khaliq et al., IEEE Access, 2025	TabNet, SwiFT, XAI, IoT	Integrates fertilizer recommendation, crop recommendation, soil analysis, and irrigation using AI and IoT.	Depends on soil/environmental data and may have limited regional generalization.	Disease-specific agrochemical recommendations are not the primary focus.	Provides insights into agricultural recommendation and decision-support methods.
10	C. Musanase et al., Agriculture, 2023	Neural Network, Rule-Based Recommendation	Uses a neural network for crop-related prediction and a rule-based system for fertilizer recommendation using soil nutrients and pH.	Developed for a specific geographic context; broader applicability requires additional environmental data.	Disease-based agrochemical recommendation is not directly addressed.	Supports the use of agricultural domain knowledge in recommendation systems.

A. Research Gap and Motivation

The literature review highlights significant progress in deep learning-based plant disease detection and agricultural recommendation systems. However, several limitations remain. Many disease classification models are evaluated using controlled datasets, which may not fully represent real-world field conditions. In addition, existing studies often focus on disease identification or agricultural recommendations separately, limiting their usefulness as integrated decision-support systems.

These limitations motivate the development of a framework that combines accurate disease classification with disease-specific agrochemical recommendation support. The proposed work uses EfficientNetB0 with transfer learning for disease detection and connects the predicted disease category to a recommendation module. This integrated approach aims to assist users in identifying crop diseases and accessing relevant crop-management guidance.

III. PROPOSED FRAMEWORK

The proposed framework integrates deep learning-based crop disease detection with agrochemical recommendation support to assist in identifying plant diseases and selecting appropriate management measures. It consists of two major components: a disease classification module and an agrochemical recommendation module. The framework accepts a plant leaf image as input and applies preprocessing operations to ensure that the image is compatible with the classification model. The preprocessed image is then analyzed using a trained EfficientNetB0 model to identify the corresponding disease category. Based on the predicted class, the recommendation component provides relevant disease-specific agrochemical information to support crop-management decisions. By combining image-based disease identification with recommendation support, the framework aims to provide a structured approach to crop disease management.

A. System Architecture

The system architecture comprises a sequence of interconnected stages, including image input, preprocessing, disease classification, and agrochemical recommendation. Initially, a user provides an image of a plant leaf showing either healthy or potentially diseased conditions. The input image is resized to the required dimensions and normalized before being passed to the EfficientNetB0 classification model. The model extracts relevant visual features, such as patterns, discoloration, and texture variations, and predicts the most likely disease category.

The predicted disease label is subsequently passed to the agrochemical recommendation module, which associates the identified condition with relevant disease-specific agrochemical information. This stage is intended to support the selection of suitable disease management measures using reliable agricultural information. The final output includes the predicted disease category and the corresponding recommendation guidance. The modular architecture allows the disease classification and recommendation components to be developed and improved independently, facilitating future extensions to additional crops and disease categories.

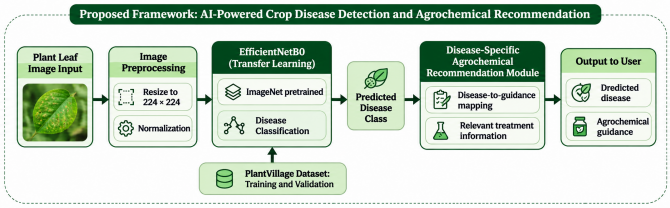


Fig. 1. System architecture of the proposed AI-powered crop disease detection and agrochemical recommendation framework.

B. Disease Detection Module

The disease detection module employs EfficientNetB0, a convolutional neural network architecture, using a transfer learning approach. The model is initialized with pretrained ImageNet weights, which provide the ability to extract general visual features such as edges, textures, shapes, and patterns from input images. These features are further adapted to recognize disease-specific characteristics present in plant leaves. The final classification layer is modified to match the 38 categories available in the PlantVillage dataset.

During classification, the preprocessed leaf image is passed through the network, where successive convolutional layers extract hierarchical representations of the input. The classification layer generates scores corresponding to the target disease categories, and the category with the highest predicted score is selected as the model output. Transfer learning enables the model to utilize previously learned visual representations, reducing the need to train the entire network from scratch and supporting efficient adaptation to the crop disease detection task.

C. Agrochemical Recommendation Module

The agrochemical recommendation module is designed to provide disease-specific guidance based on the disease category predicted by the classification module. The predicted label serves as an input to the recommendation component, which associates the identified disease with relevant agrochemical information and potential disease management measures. This integration connects image-based disease identification with actionable agricultural guidance, helping users understand possible treatment options for the detected condition.

The recommendation process should consider crop-specific requirements, disease characteristics, and reliable agricultural information to ensure that the guidance is relevant and appropriate. Any suggested agrochemical must be verified against the product label, recommended dosage, safety precautions, and applicable agricultural guidelines. The module is intended to support crop-management decisions and should not replace the advice of qualified agricultural experts. Its effectiveness and safety depend on the accuracy and reliability of the underlying recommendation information.

IV. Methodology

The proposed methodology is designed to facilitate the automated identification of crop diseases from plant leaf images and support disease-specific agrochemical recommendations. The overall framework consists of four major stages: dataset preparation, image preprocessing, transfer learning-based model training, and performance evaluation. The PlantVillage dataset is utilized to train and validate the disease classification model, while EfficientNetB0 is adopted as the feature extraction and classification backbone. The model learns discriminative visual patterns associated with different plant health conditions, including variations in leaf color, texture, and visible disease symptoms.

Initially, the dataset is organized into the required training, validation, and testing subsets to support model development and performance assessment. The input leaf images are then preprocessed to ensure a consistent representation before being passed to the neural network. During model training, transfer learning is employed by utilizing the pretrained EfficientNetB0 architecture and adapting its classification capability to the target plant disease categories. The model parameters are optimized using a suitable loss function and optimization algorithm, while validation performance is monitored to assess learning progress and identify the best-performing model.

Following the classification stage, the predicted disease category is associated with the corresponding disease-specific information in the agrochemical recommendation component. This component is intended to provide relevant disease management guidance based on verified agricultural information. Finally, the classification model is evaluated using appropriate performance metrics to analyze its effectiveness in distinguishing between healthy and diseased plant leaves. The overall workflow of the proposed methodology is illustrated in Fig. 3.

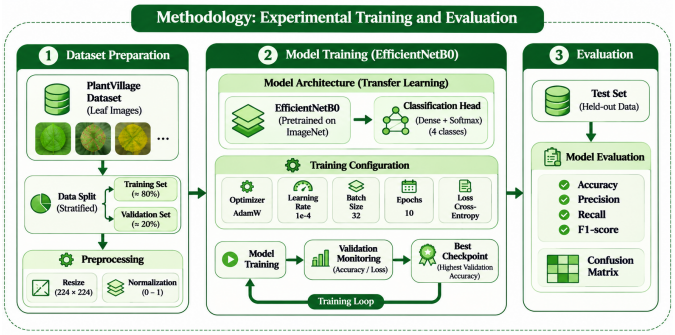


Fig. 2. Experimental workflow for training and evaluating the EfficientNetB0-based crop disease classification model.

A. Dataset Description

The PlantVillage dataset is utilized to train and validate the proposed crop disease detection model. It contains 54,305 images of healthy and diseased plant leaves distributed across 38 categories, representing different crop varieties and disease conditions. The dataset provides labeled images that enable the model to learn visual characteristics associated with specific plant diseases. For model development, the dataset is divided into training, validation, and testing subsets. The training subset is used to learn model parameters, while the validation subset is used to monitor performance during training and support model selection. The testing subset is reserved for independent evaluation of the trained model.

B. Image Preprocessing

Image preprocessing is performed to ensure that the input data are compatible with the EfficientNetB0 architecture. All input leaf images are resized to 224×224 pixels, providing a uniform spatial dimension for model processing. The pixel values are normalized to maintain a consistent input representation and support stable model training. These preprocessing operations reduce variations in image dimensions and input scale, allowing the model to focus on relevant visual features such as leaf texture, discoloration, and disease-related patterns. The resulting preprocessed images are supplied to the classification network.

C. EfficientNetB0-Based Classification

EfficientNetB0 is adopted as the backbone architecture for plant disease classification because of its efficient feature extraction capability and comparatively compact model structure. The network utilizes pretrained ImageNet weights to extract general visual features, including edges, textures, and patterns, from the input leaf images. These pretrained features provide a useful starting point for identifying disease-specific characteristics. The final classification layer is modified to accommodate the 38 categories of the PlantVillage dataset. Through transfer learning, the model adapts its learned representations to the plant disease classification task, enabling it to distinguish between healthy leaves and different disease conditions.

D. Model Training

The model is trained using the prepared training dataset, while the validation dataset is used to monitor its performance throughout the training process. During each training iteration, the input images are passed through the network to obtain class predictions. The predicted outputs are compared with the corresponding ground-truth labels using the classification loss function. An optimization algorithm updates the model parameters to minimize the loss and improve classification performance. Validation accuracy and loss are monitored after each epoch to assess the model's learning behavior and identify potential overfitting. The model checkpoint corresponding to the best validation performance is selected for subsequent evaluation.

E. Performance Evaluation

The performance of the disease detection model is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model's ability to correctly identify plant disease categories.

V. Results and Discussion

The proposed crop disease detection model was evaluated using the PlantVillage dataset. The dataset contains 54,305 leaf images distributed across 38 disease and healthy-leaf categories. The images were divided into training, validation, and testing subsets containing 43,444, 5,430, and 5,431 images, respectively.

The EfficientNetB0 model was trained using transfer learning for 10 epochs. The model achieved its best recorded validation accuracy of 99.8343% at Epoch 9, with a validation loss of 0.0053. At Epoch 10, the training accuracy was 99.73%, while the validation accuracy was 99.78%. These results indicate that the model learned discriminative features for identifying crop leaf conditions within the PlantVillage dataset.

As shown in Fig. **??**, the model achieved its best validation accuracy of 99.8343% at Epoch 9.

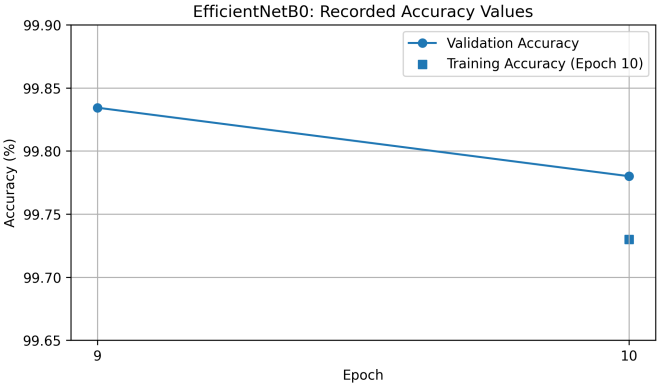


Fig. 3. Recorded Training and Validation Accuracy of EfficientNetB0

A. Performance Analysis

The validation accuracy demonstrates the effectiveness of the EfficientNetB0 architecture for image-based crop disease classification. The use of a pretrained network allows the model to leverage previously learned visual features, reducing the need to learn all image representations from scratch.

The difference between the training and validation accuracy at the final epoch is relatively small, suggesting that the model maintained consistent performance on the validation data during training. However, validation accuracy alone does not establish the model's ability to generalize to unseen field images. Further evaluation using the independent test set and images captured under real agricultural conditions is required.

B. Discussion

The experimental results show the potential of EfficientNetB0 for automated crop disease detection. The model can support the identification of disease-related leaf patterns and provide a foundation for an integrated crop disease detection and agrochemical recommendation framework.

The proposed framework is intended to connect disease classification with disease-specific recommendation information. However, the current experimental results primarily validate the image classification component. The effectiveness and safety of agrochemical recommendations require separate validation using reliable agricultural sources and expert guidance.

Although the model achieved high validation accuracy on PlantVillage, the dataset consists largely of controlled-condition images. Variations in lighting, background, camera quality, crop variety, and disease severity may affect performance in real-world agricultural environments. Therefore, field-level validation is an important direction for future work.

TABLE II
TRAINING AND VALIDATION PERFORMANCE OF EFFICIENTNETB0

Parameter	Value
Dataset	PlantVillage
Number of Classes	38
Training Images	43,444
Validation Images	5,430
Test Images	5,431
Input Image Size	224×224
Number of Epochs	10
Best Validation Accuracy	99.8343%
Best Validation Loss	0.0053
Best Epoch	9

VI. CONCLUSION AND FUTURE WORK

This paper presents an AI-powered crop disease detection and agrochemical recommendation framework aimed at supporting early disease identification and improving agricultural decision-making. The proposed framework combines a deep learning-based image classification model with a disease-specific recommendation component. EfficientNetB0, a convolutional neural network architecture, was employed through transfer learning to identify healthy and diseased plant leaves from input images. The classification model was evaluated using the PlantVillage dataset, which consists of 38 categories representing different crop and disease conditions. The framework is designed to provide a structured approach to disease identification and to support the selection of suitable disease management measures.

The experimental results demonstrate the effectiveness of the EfficientNetB0 model under the experimental conditions of the PlantVillage dataset. The model achieved a best validation accuracy of 99.8343% at Epoch 9, with a corresponding validation loss of 0.0053. These results indicate that transfer learning can provide a strong foundation for automated plant disease classification when sufficient labeled image data are available. The high validation performance also highlights the potential of deep learning techniques to reduce the dependence on manual visual inspection and assist in the timely identification of crop diseases. Nevertheless, the reported validation results should be interpreted within the limitations of the dataset, as images captured in controlled environments may differ significantly from those collected in actual agricultural fields.

Although the proposed framework integrates disease detection with agrochemical recommendation, the reliability and practical effectiveness of the recommendation component require further validation. Agrochemical selection depends on several factors, including the identified disease, crop variety, disease severity, environmental conditions, and applicable agricultural guidelines. Therefore, recommendations must be based on verified agricultural information and should be reviewed by qualified agricultural experts before practical deployment. The framework should be considered a decision-support system rather than a replacement for professional agronomic advice.

Future work will focus on evaluating the trained model using an independent test dataset and collecting field images under diverse environmental conditions, including variations in illumination, background, leaf orientation, and disease severity. Such evaluation will help determine the model's generalization capability beyond the PlantVillage dataset. Further improvements may include incorporating additional crop varieties and disease categories, optimizing the model for resource-constrained devices, and developing a user-friendly interface for farmers. The agrochemical recommendation component will also be enhanced by integrating reliable agricultural databases, crop-specific treatment guidelines, and expert-validated information. In addition, explainable AI techniques may be explored to highlight the image regions contributing to disease predictions, thereby improving transparency and user confidence. These developments can help transform the proposed framework into a more reliable, accessible, and practical tool for precision agriculture and sustainable crop management.

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